The first positive Mittag–Leffler zero near order one

Principia Math In Progress

21 September 2026

Task 1: MathDB 361626. For $a > 0$, put

$$f_a(t) = E_{a,1}(-t^a) = \sum_{k=0}^{\infty} \frac{(-1)^k t^{ak}}{\Gamma(ak+1)}, \qquad t \geq 0,$$

and let $Z_1(a)$ be its smallest positive zero, when one exists. Conjecture 3(b) of Chaudhary, Diethelm and Hashemishahraki [1] proposes, as $a \downarrow 1$,

$$Z_1(a) = (c \log(a-1) + d)(1 + o(1)), \qquad c \approx -0.275, \quad d \approx 17.54.$$

The following theorem disproves the proposed leading coefficient and gives the asymptotic with an additive error tending to zero.

Theorem 1. *For all sufficiently small $\varepsilon > 0$, $f_{1+\varepsilon}$ has a smallest positive zero. With $L = \log(1/\varepsilon)$,*

$$\boxed{Z_1(1+\varepsilon) = L + \log L + o(1).}$$

In particular $Z_1(1+\varepsilon)/L \to 1$.

The conjectured leading coefficient is therefore forced to be $c = -1$, far from its stated numerical fit. This is a disproof of part (b), and hence of the conjunction recorded in MathDB. The global positivity bound in part (a) and the endpoint expansion at $a = 2$ in part (c) are not settled here. A constant d is invisible at the conjecture's multiplicative $1 + o(1)$ precision; no contradiction is based on d.

A classical integral representation

Throughout the proof $a = 1 + \varepsilon$, $0 < \varepsilon < 1/4$, and $\theta = \pi\varepsilon$. Define the positive density

$$K_\varepsilon(r) = \frac{\sin\theta}{\pi} \frac{r^{a-1}}{(r^a-1)^2 + 2r^a(1-\cos\theta)}, \qquad r > 0.$$

The standard residue and branch-cut representation is

$$f_a(t) = \frac{2}{a} e^{t\cos(\pi/a)} \cos(t\sin(\pi/a)) - \int_0^\infty e^{-tr} K_\varepsilon(r)\, dr \quad (t > 0). \tag{1}$$

For completeness, termwise Laplace transformation of the defining series for real $s > 1$ gives

$$\int_0^\infty e^{-st} f_a(t)\, dt = \sum_{k\geq 0} (-1)^k s^{-ak-1} = \frac{s^{a-1}}{s^a + 1}.$$

Use the principal branch, cut along the negative real axis. There are two simple poles at $e^{\pm i\pi/a}$; their inverse-Laplace residues are $a^{-1}\exp(te^{\pm i\pi/a})$. On the upper side of the cut,

$$\left.\frac{s^{a-1}}{s^a+1}\right|_{s=-r+i0} = \frac{r^{a-1}e^{i\theta}}{1-r^a e^{i\theta}}, \qquad \frac{1}{\pi}\operatorname{Im}\frac{r^{a-1}e^{i\theta}}{1-r^a e^{i\theta}} = K_\varepsilon(r).$$

Deforming the inversion contour to the cut consequently gives the negative integral in (1). The small-circle contribution is $O(\rho^a)$ as its radius $\rho \downarrow 0$; the large arcs vanish for $t > 0$ since the transform is $O(1/s)$ and the exponential decays on the left arcs. The cut integral is absolutely convergent at both endpoints.

The exact total mass, obtained with $u = r^a$, is

$$\int_0^\infty K_\varepsilon(r)\,dr = \frac{\sin\theta}{\pi a}\int_0^\infty \frac{du}{(u-\cos\theta)^2+\sin^2\theta} = \frac{\pi-\theta}{\pi a} = \frac{2}{a} - 1. \tag{2}$$

The antiderivative used here is $(\pi a)^{-1}\arctan((u-\cos\theta)/\sin\theta)$. The representation and the exponential/algebraic decomposition are classical; see, for example, [2, 3, 4]. Our argument below supplies the needed uniform bounds and identifies the *first* zero in the simultaneous limit.

Uniform expansion in a logarithmic window

Lemma 1. *Uniformly for $L/4 \le t \le 3L/2$, as $\varepsilon \downarrow 0$,*

$$f_a(t) = e^{-t}(1+o(1)) - \varepsilon\Gamma(a)t^{-a}(1+o(1)). \tag{3}$$

The two errors in this formula are separately uniform relative errors.

Proof. All constants in the following estimates can be chosen independently of ε and t in this window. Put $h = t^{-2}$, so eventually $h < 1/2$. Write $D(r) = 1 - 2r^a\cos\theta + r^{2a}$. For $0 \le r \le 1/2$, $D(r) \ge (1-r^a)^2 \ge 1/4$ and $D(r)^{-1} = 1 + O(r^a)$. Thus

$$\begin{aligned}\int_0^{1/2} e^{-tr}K_\varepsilon(r)\,dr &= \frac{\sin\theta}{\pi}\left\{\Gamma(a)t^{-a} + O(t^{-2a}) + O(t^{-a}(1+t)e^{-t/2})\right\}\\ &= \varepsilon\Gamma(a)t^{-a}(1+o(1)).\end{aligned}$$

Here the $O(t^{-2a})$ bound follows by extending the error integral to infinity; its factor $\Gamma(2a)$ is uniformly bounded. For the main gamma integral the omitted tail is bounded, after $u = tr$, by $t^{-a}\int_{t/2}^\infty (1+u)e^{-u}\,du$.

Next, the unweighted K_ε mass outside $[1/2, 3/2]$ is $O(\varepsilon)$. The lower part follows from $D \ge 1/4$; on the upper part $D \ge (r^a-1)^2 \ge r^{2a}/9$, and integration of $O(\varepsilon r^{-a-1})$ gives the assertion. On $[1/2, 3/2]$, r^{a-1} is bounded and the mean value theorem gives $|r^a - 1| \ge C_0|r-1|$ with a fixed $C_0 > 0$. Consequently

$$K_\varepsilon(r) \le \frac{C\varepsilon}{(r-1)^2} \quad (r \ne 1).$$

Combining this estimate with (2) yields

$$\int_{1-h}^{1+h} K_\varepsilon(r)\,dr = 1 + O(\varepsilon) + O(\varepsilon/h) = 1 + o(1),$$

since $\varepsilon t^2 \to 0$ uniformly. On this central interval $e^{-tr} = e^{-t}(1 + O(th))$; hence its weighted contribution is

$$\int_{1-h}^{1+h} e^{-tr} K_\varepsilon(r)\, dr = e^{-t}(1 + o(1)).$$

On the rest of $[1/2, 3/2]$ the contribution is at most $C\varepsilon h^{-2} e^{-t/2} = C\varepsilon t^4 e^{-t/2}$. Dividing by e^{-t} bounds it by

$$C\varepsilon(3L/2)^4 e^{3L/4} = O(L^4 \varepsilon^{1/4}) = o(1).$$

The remaining tail $r \geq 3/2$ contributes at most $C\varepsilon e^{-3t/2} = o(e^{-t})$.

Finally, $\cos(\pi/a) = -1 + O(\varepsilon^2)$ and $\sin(\pi/a) = O(\varepsilon)$. Since $\varepsilon t \to 0$, the residue term in (1) is $2e^{-t}(1 + o(1))$. Subtracting the central contribution $e^{-t}(1 + o(1))$ and the small-r contribution proves (3). □

Excluding earlier zeros

Lemma 2. *For all sufficiently small $\varepsilon > 0$, $f_a(t) > 0$ for $0 \leq t \leq L/4$.*

Proof. We compare directly with $f_1(t) = e^{-t}$. For $1 \leq b \leq a$ and $k \geq 1$,

$$\frac{\partial}{\partial b} \frac{t^{bk}}{\Gamma(bk+1)} = \frac{k t^{bk}}{\Gamma(bk+1)} (\log t - \psi(bk+1)).$$

The standard gamma bounds $\Gamma(bk+1) \geq k!$ and $0 < \psi(bk+1) \leq \log(bk+1) \leq 2k$ apply. The first uses monotonicity of Γ on $[2, \infty)$; the digamma upper bound also follows from Jensen's inequality for the logarithm under a gamma density. These estimates justify differentiating the series on compact parameter and t intervals.

For $1 \leq t \leq L/4$, eventually $t^\varepsilon \leq 2$. Put $T = t^b \leq 2t$. Summing absolute values of the derivatives gives

$$\begin{aligned} |\partial_b f_b(t)| &\leq \sum_{k \geq 1} \frac{(k \log t + 2k^2) T^k}{k!} \\ &= (T \log t + 2T^2 + 2T) e^T \leq C(1+t)^3 e^{2t}. \end{aligned}$$

The fundamental theorem of calculus therefore implies

$$\frac{|f_a(t) - e^{-t}|}{e^{-t}} \leq C\varepsilon(1+L)^3 e^{3L/4} = O(\varepsilon^{1/4}(1+L)^3) = o(1).$$

For $0 < t \leq 1$, use $k t^{bk} |\log t| \leq 1/(b\mathrm{e}) \leq 1/\mathrm{e}$. The derivative series is then uniformly bounded by $\sum_{k \geq 1} (1/\mathrm{e} + 2k^2)/k!$, giving $|f_a(t) - e^{-t}| = O(\varepsilon)$ uniformly. Since $e^{-t} \geq e^{-1}$ there and $f_a(0) = 1$, positivity follows on the entire interval. □

Locating the first zero

Fix $0 < \delta < 1/2$. On $[L/4, (1-\delta)L]$ consider the ratio of the second leading term in (3) to the first:

$$R_\varepsilon(t) = \varepsilon \Gamma(a) e^t t^{-a}.$$

For large L this is increasing throughout the interval because its logarithmic derivative is $1 - a/t > 0$. At its upper endpoint it is $O(e^{-\delta L} L^{-a}) = o(1)$. Lemma 1 therefore gives positivity throughout this

interval. Together with Lemma 2 this rules out every zero up to $(1-\delta)L$. At $t=(1+\delta)L$, the ratio is $\Gamma(a)e^{\delta L}((1+\delta)L)^{-a}\to\infty$, so (3) gives a negative value. Continuity now supplies a zero, and the zero set has a least positive element since it is closed and bounded away from zero. It follows that

$$(1-\delta)L < Z_1(a) < (1+\delta)L.$$

As δ was arbitrary, $Z_1(a)/L\to 1$.

At this first zero, both relative errors in (3) tend to zero. Taking logarithms of the equality of the two terms gives

$$Z_1(a) = L + a\log Z_1(a) - \log\Gamma(a) + o(1).$$

Now $\log Z_1(a) = \log L + o(1)$, $\varepsilon\log Z_1(a) = o(1)$, and $\log\Gamma(a) = o(1)$. This proves Theorem 1. □

Verification and scope

The proof is analytic and does not infer uniformity or first-zero minimality from numerical samples. The accompanying verifier checks the branch-cut algebra and mass antiderivative symbolically. It also performs supplemental 100-digit power-series calculations for five values of ε from 10^{-2} to 10^{-20}, checks the predicted sign transition, and compares two values against independent quadrature of (1). These computations are not interval certificates. Review is by the authoring agent only. The complete current preprint and the complete author-posted text of its journal final were read; the bounded priority audit is documented separately and does not certify exhaustive novelty.

Four proposed follow-up tasks

Task 2: Determine the next term after $L+\log L$, including whether it is $(\log L-2)/L+o(1/L)$.

Task 3: Give explicit constants and an explicit ε_0 in a two-sided error bound for the first-zero asymptotic.

Task 4: Prove or refute the source's global bound $f_a(t)>0$ for $1<a\le 2$ and $0\le t\le 1.559$.

Task 5: Derive a closed expression for the coefficient of $2-a$ in $Z_1(a)$ at $a=2$ and rigorously compare it with the source's fit.

References

[1] R. Chaudhary, K. Diethelm and S. Hashemishahraki, On the separation of solutions to fractional differential equations of order $\alpha\in(1,2)$, *Applied Numerical Mathematics* **203** (2024), 84–96, doi:10.1016/j.apnum.2024.05.020. Current preprint: arXiv:2401.14771v3. Conjecture 3(b), journal p. 95; preprint p. 17.

[2] J. W. Hanneken, B. N. N. Achar and D. M. Vaught, An alpha-beta phase diagram representation of the zeros and properties of the Mittag-Leffler function, *Advances in Mathematical Physics* (2013), Article 421685, doi:10.1155/2013/421685.

[3] R. B. Paris, Exponential asymptotics of the Mittag–Leffler function, *Proceedings of the Royal Society A* **458** (2002), 3041–3052, doi:10.1098/rspa.2002.0977.

[4] R. B. Paris, Asymptotics of the Mittag–Leffler function $E_a(z)$ on the negative real axis when $a\to 1$, arXiv:2005.05737v2 (2021). This paper restricts $a<1$; its first-zero counterpart is not asserted.